

Inferring and validating the hidden from the observed



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Motivation

- Animal movement is influenced by underlying behaviour

<https://www.movebank.org/cms/downloads/video-1280.mp4>

- Inferred behaviours of animals from GPS movement data are rarely validated

Source: movebank

Goal

Validate HMM-inferred behaviours
of seabird from movement data



Figure 1: *red listed seabirds in the UK*

Study species

- Visually tracked in the UK by Joint Nature Conservation Committee between 2009 - 2011

Movement data: GPS coordinates of the boat

Behavioural data: taken by the observer on the boat

Arctic tern



Common tern



Roseate tern



Sandwich tern



Figure 2: *Species of terns in the UK*

Study site

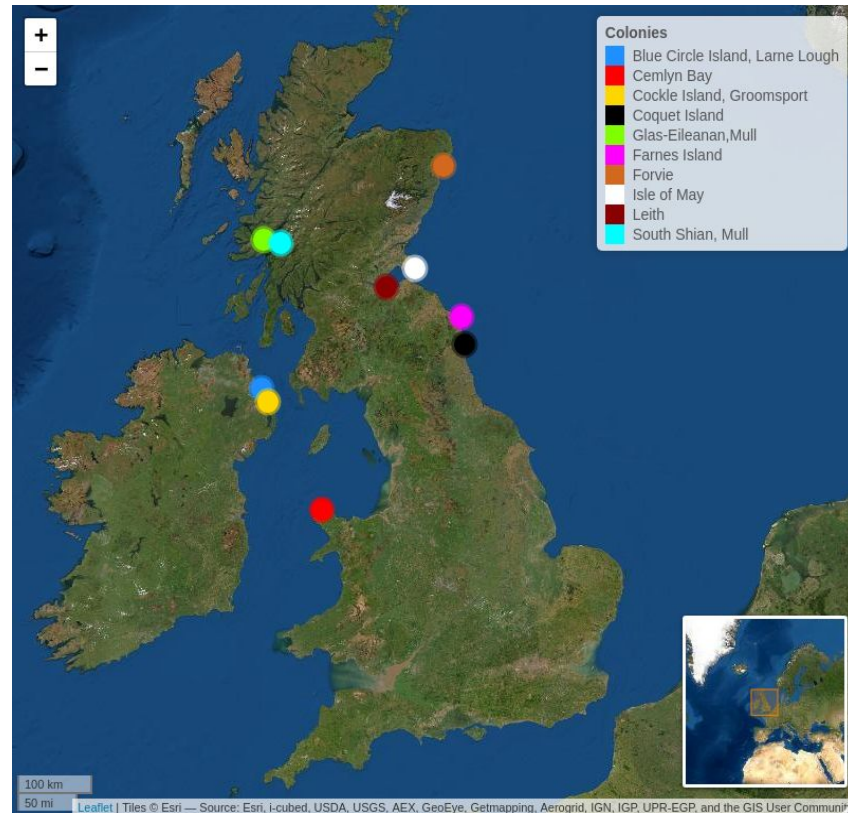


Figure 3: Study site consisting of 9 tern breeding colonies in the UK

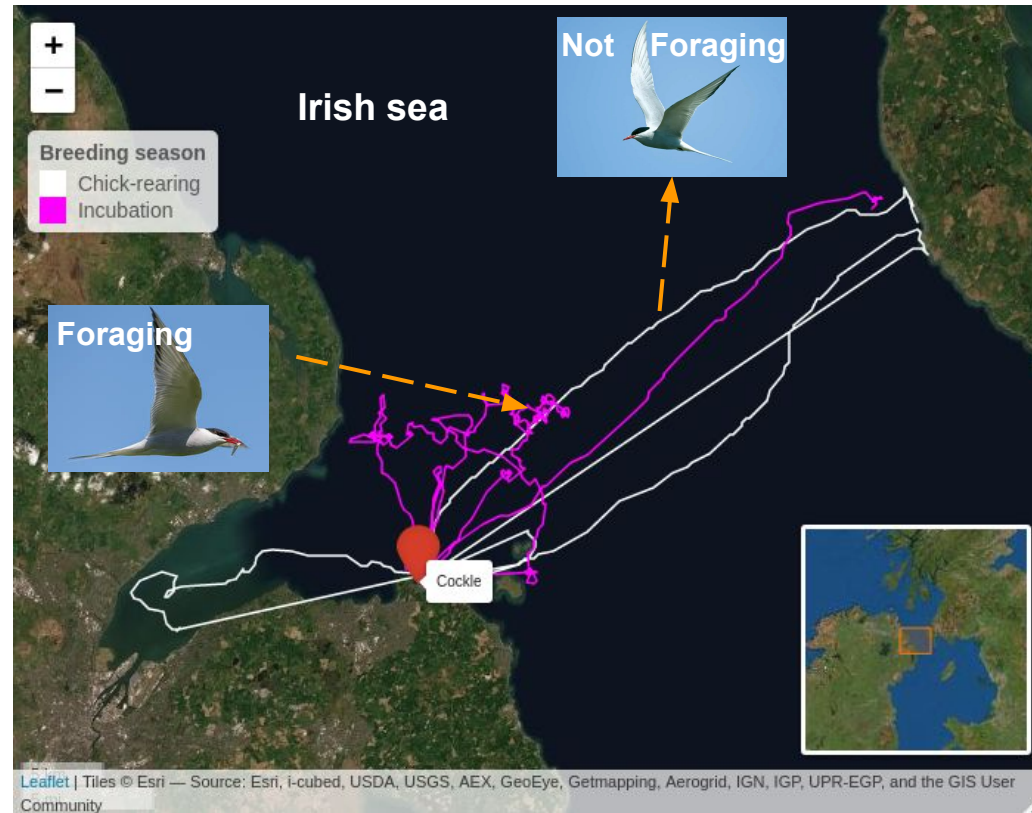


Figure 4: Visual tracking (boat movement) data of sandwich terns at Cockle Island colony ($54^{\circ}67'N, 5^{\circ}61'W$), Northern Ireland, 2010

Hidden Markov model

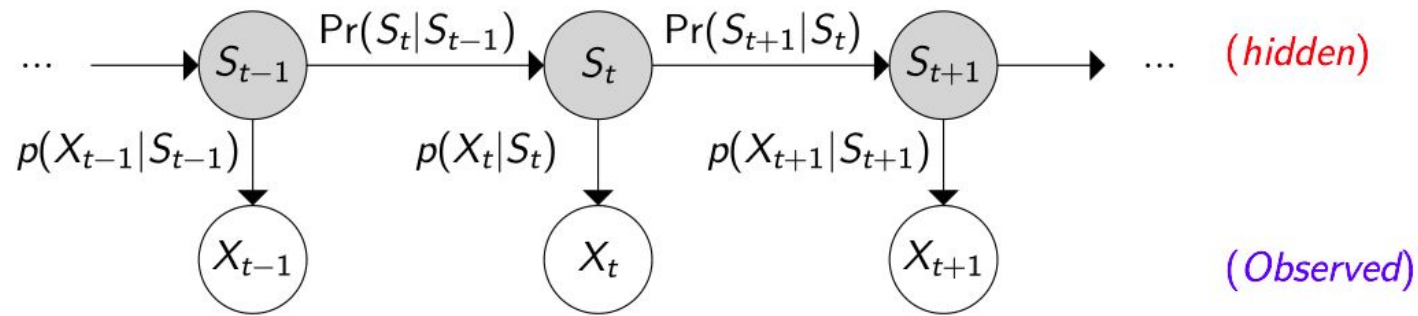


Figure 5: Representation of a HMM

- Tern behavioural states: foraging and not-foraging
- Step length (gamma) and turning angle (von Mises) obtained from proxy (boat) movement data of terns

Result

Table 1: Summary of visual tracking data of terns across the breeding colonies during the chick-rearing period.

Chick-rearing 2010			
Colony	Tern species	No. of tracks	Total tracking seconds
Isle of May	Arctic	5	3292
South Shian	Common	6	14272
Coquet	Roseate	1	2984
Cockle	Sandwich	5	13014

Table 2: Validation result of HMM fitted to visual tracking data of terns during the chick-rearing breeding season. PPV = positive predictive value (foraging), NPV = negative predictive value (not-foraging), TPR = true positive rate.

Chick-rearing, 2010		Validation metrics				
Colony	Species	PPV	NPV	TPR	F1-score	log-loss
Isle of May	Arctic	0.8641	0.7170	0.7099	0.7794	7.5165
South Shian	Common	0.7026	0.9668	0.8995	0.7889	3.9145
Coquet	Roseate	0.6807	0.9030	0.8372	0.7508	6.6749
Cockle	Sandwich	0.8776	0.9531	0.8602	0.8688	2.3079

Result

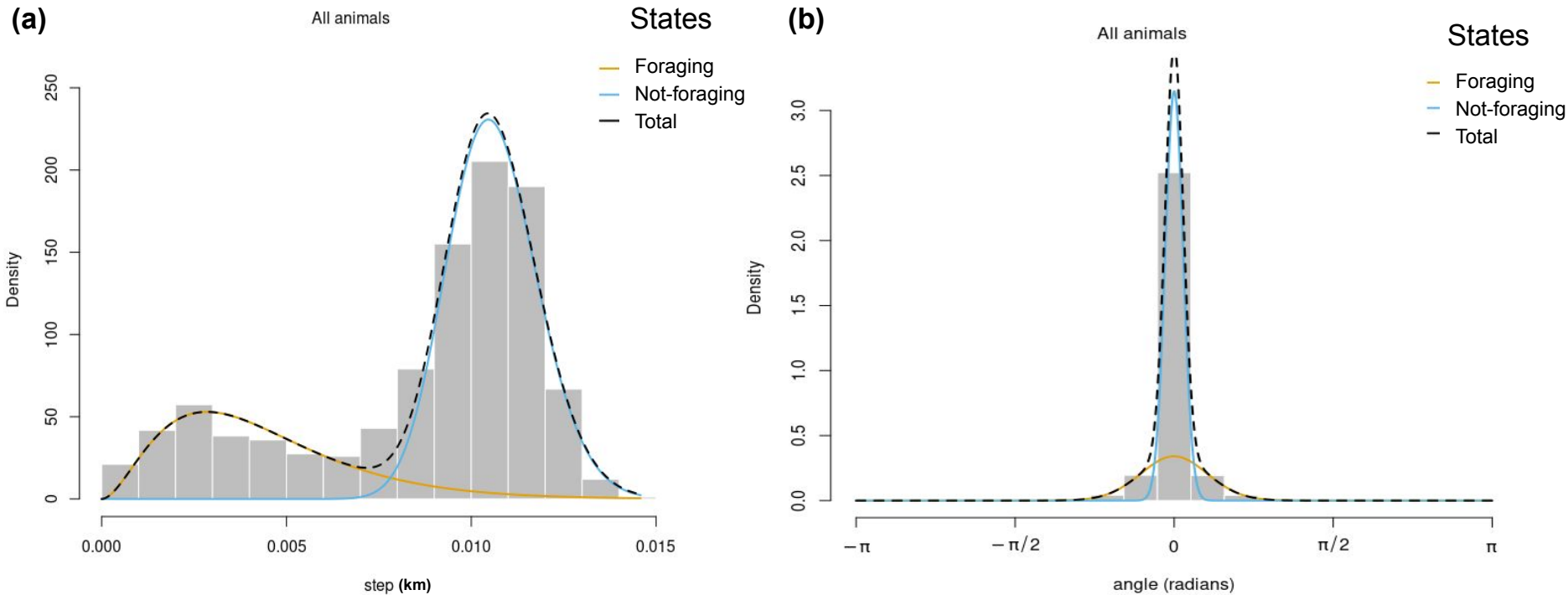
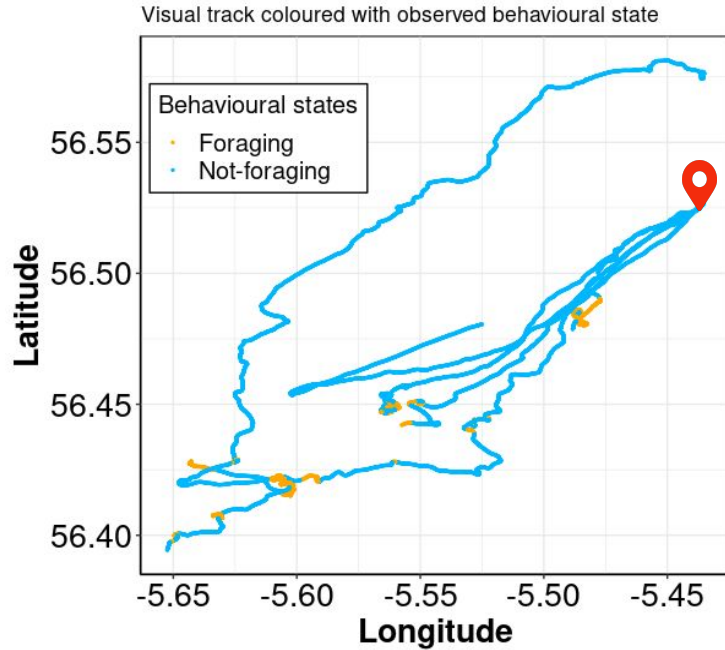


Figure 6: Histograms of the (a) proxy step length and (b) turning angle distributions of 6 visually-tracked common tern in **South Shian colony** during chick-rearing period, 2011. Lines represent the HMM fitted state-dependent distributions, and are coloured according to decoded behavioural state: **foraging** and **not-foraging**

Result

(a)



(b)

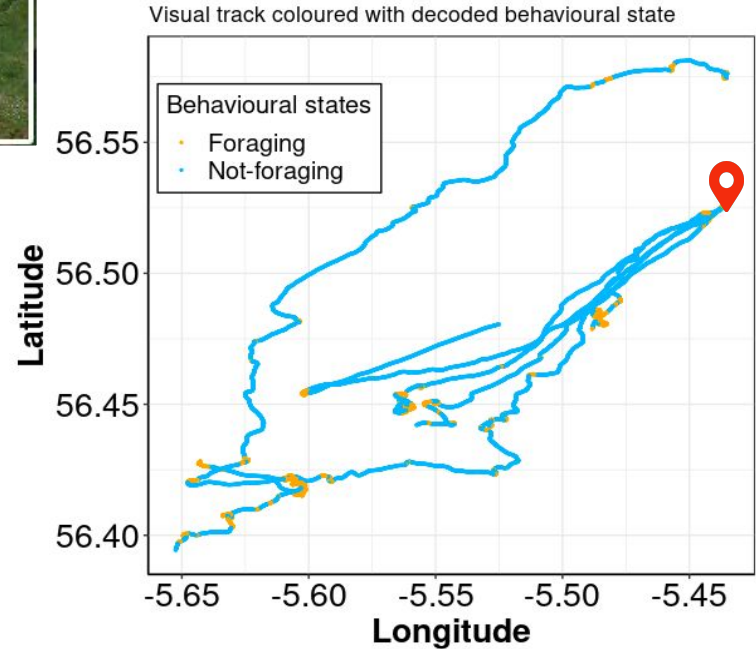


Figure 7: proxy (boat) movement data of 6 visually tracked common tern coloured with (a) observed and (b) decoded behavioural states: *foraging* and *not-foraging*, in **South Shian colony** (📍) Mull, Scotland, during chick-rearing period, 2011.

Take home

- Visual tracking is a suitable proxy to terns GPS movement data
- HMMs can correctly infer behaviours from GPS movement data provided key assumptions are not violated

Ongoing Work

Assess the effect of different sampling frequency of GPS movement data on inferred behaviours

References

- Hidden Markov Models for Time Series; An Introduction Using R

By: Walter Zucchini, Iain L. MacDonald and Roland Langrock.

- Animal Movement; Statistical Models for Telemetry Data

By: Mevin B. Hooten, Devin S. Johnson, Brett T. McClintock and Juan M. Morales.

Thank you for your attention



Questions?